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Progress Report Number: 4

## Work Date/Hours Logs

| Date         | Hours | Description of work done                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |
|--------------|-------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Mar 15, 2026 | 6     | <b>DDoS Detection Troubleshooting &amp; Threshold Refinement</b><br>Debugged inconsistent DDoS detection triggers; analyzed why Rule2 (latency-based) and Rule3 (error-rate based) were not firing reliably during attack simulations. Tested various traffic patterns and adjusted thresholds in classifier.py. Enhanced ddos_simulator.py to generate more realistic attack patterns with 3000 threads and varied endpoints. Verified Rule1 (resource saturation) works consistently but identified need for more sensitive detection parameters for comprehensive DDoS classification.                                                                                                           |
| Mar 16, 2026 | 6     | <b>Zero-Baseline Fix &amp; Lab Automation Suite Development</b><br>Fixed critical zero-baseline handling in baseline.py for auth_failures anomaly detection - implemented logic to treat any positive value as anomalous when baseline mean is zero, enabling reliable brute force detection. Developed comprehensive lab automation suite: lab-startup.sh (complete setup with permanent golden baseline), quick-start.sh (fast daily startup), reset-baseline.sh (baseline recreation), and simulate-attacks.sh (automated attack testing). Solved baseline persistence problem - eliminated need for daily baseline recreation through golden baseline system with Kubernetes ConfigMap storage. |
| Total: 12    |       |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |

## Summary Description of Work Done During This Reporting Period

This reporting period focused on **resolving critical detection issues** and **implementing comprehensive lab automation** to enhance research efficiency and reproducibility.

Work Completed:

*1. DDoS Detection Debugging & Enhancement (Mar 15)*

- **Root Cause Analysis:** Identified that Rule2 (user impact) and Rule3 (service failure) had overly conservative thresholds causing missed detections during legitimate attack simulations.
- **Threshold Refinement:** Adjusted detection sensitivity in classifier.py to balance false positive reduction with attack detection accuracy.

- **Simulation Enhancement:** Upgraded ddos\_simulator.py to generate more realistic attack patterns using 3000 threads with random endpoint targeting, better mimicking real-world DDoS attacks.
- **Validation Testing:** Confirmed Rule1 (resource saturation) consistently detects high-volume attacks, documented specific conditions required for Rule2 and Rule3 activation.

## *2. Zero-Baseline Anomaly Detection Fix (Mar 16)*

- **Critical Bug Resolution:** Fixed fundamental issue in baseline.py where metrics with zero baseline means (like auth\_failures) were not properly flagged as anomalous when positive values appeared.
- **Enhanced Detection Logic:** Implemented logic to assign 1,000,000% deviation for any positive value when baseline mean is zero, ensuring immediate anomaly detection.
- **Brute Force Enablement:** This fix directly enables Rule4 (brute force detection) to function correctly, as authentication failures typically have zero baseline under normal conditions.

## *3. Comprehensive Lab Automation Suite (Mar 16)*

- **Baseline Persistence Solution:** Developed golden baseline system that eliminates daily baseline recreation requirement - baseline is created once and persists across pod restarts via Kubernetes ConfigMaps.
- **Workflow Automation:** Created four specialized scripts:
  - lab-startup.sh: Complete lab initialization with permanent baseline management
  - quick-start.sh: Fast 2-3 minute daily startup (vs 15+ minute manual process)
  - reset-baseline.sh: Baseline recreation tool for system configuration changes
  - simulate-attacks.sh: Automated attack testing with real-time monitoring and data collection
- **Research Efficiency:** Transformed manual, time-consuming lab setup into automated, reproducible workflow optimized for daily research activities.

## *4. Documentation & Methodology Enhancement*

- **Comprehensive Documentation:** Created detailed README.md with usage guides, troubleshooting, and research workflow recommendations.
- **Reproducible Experiments:** Automation ensures consistent baseline across all experiments, critical for valid comparative analysis in thesis evaluation.
- **Data Collection:** Implemented automated timestamped result collection in JSON format for systematic analysis.
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### Next Steps:

- **Threshold Optimization:** Fine-tune Rule2 and Rule3 detection parameters based on automated testing results
- **Comprehensive Evaluation:** Use automation suite to run systematic experiments for thesis analysis
- **Dashboard Integration:** Display real-time incident classifications in React frontend

## Repo Check-in of Implementation Completed

The following files were **modified or created** and checked into GitHub:

### Modified Files (Zero-Baseline Fix):

- **backend/src/baseline.py** – Fixed zero-baseline handling in `get_current_deviation()` method to properly detect anomalies when baseline mean is zero (critical for `auth_failures` detection)

### New Files (Lab Automation Suite):

- **lab-startup.sh** – Complete lab setup script with permanent golden baseline management and Prometheus stack installation
- **quick-start.sh** – Fast daily startup script optimized for regular research workflow (2-3 minute startup)
- **reset-baseline.sh** – Baseline recreation tool for system configuration changes with automatic backup
- **simulate-attacks.sh** – Comprehensive attack simulation suite with real-time monitoring and automated data collection
- **README.md** – Complete documentation for automation suite including usage guides and troubleshooting

### Enhanced Files (DDoS Troubleshooting):

- **scripts/ddos\_simulator.py** – Enhanced to generate more realistic attack patterns with 3000 threads and random endpoint targeting

**All automation scripts tested and verified** in Kubernetes environment. The system now provides **reproducible research workflow** with **permanent baseline persistence** and **automated experiment execution**.

## AI Use Section

| AI Tool Name    | Version, Type Account       | Specific feature for which the AI tool was used                        | Value Addition (What value did you add over and above what AI did for you?)                                                                                                                                    |
|-----------------|-----------------------------|------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| DeepSeek        | DeepSeek Chat, Free Account | <b>DDoS threshold debugging</b> and attack simulation pattern analysis | Systematically tested AI-suggested thresholds against real attack data, validated detection accuracy through multiple simulation runs, and documented specific conditions for each rule activation.            |
| Claude Sonnet 4 | Claude.ai, Free Account     | <b>Zero-baseline anomaly detection</b> logic design                    | Adapted AI-provided logic to handle specific edge cases in SRE metrics, integrated seamlessly with existing Prometheus query structure, and validated against real auth_failures scenarios.                    |
| Claude Sonnet 4 | Claude.ai, Free Account     | <b>Lab automation architecture</b> and script structure                | Designed comprehensive automation suite specifically for SRE security research workflow, integrated with existing Kubernetes infrastructure, and optimized for academic research reproducibility requirements. |
| Claude Sonnet 4 | Claude.ai, Free Account     | <b>Golden baseline persistence</b> system design                       | Implemented Kubernetes ConfigMap integration for baseline storage, created backup and recovery mechanisms, and designed system to handle both first-time setup and daily research workflows.                   |
| Claude Sonnet 4 | Claude.ai, Free Account     | <b>Attack simulation automation</b> and data collection framework      | Enhanced AI-provided scripts with real-time monitoring, automated result collection with timestamps, and integrated with existing detection framework for comprehensive experiment workflow.                   |